Experiment 1

Aim: Practice of LEX/YACC of compiler writing

Theory:

LEX is a tool used to generate lexical analyzers, the first phase of a compiler. It takes a source file containing regular expressions and their associated actions, written in a file with a .l extension, and translates it into a C program (lex.yy.c) that recognizes those patterns in an input stream. Each recognized pattern is called a token, and the generated scanner reads characters from the input, groups them into these tokens based on the rules defined, and passes them on to later stages of the compiler such as the parser. A LEX program is organized into three sections separated by %%: the definitions section for declarations and options, the rules section containing the actual patterns and actions, and the user subroutines section for supporting C functions like main(). Internally, LEX follows the longest-match rule, meaning it always picks the longest string that matches any rule, and if two rules match the same length, the one defined first in the file wins. This makes LEX efficient for building scanners without having to hand-write the matching logic in raw C.

Practice Questions

1. Write a program to construct a small calculator that can add, subtract, multiply and divide in LEX.

Code:

```
%{
#include <stdio.h>
#include <stdlib.h>
float num1, num2, result;
char op;
%}
%%
[0-9]+"."[0-9]+    { num2 = atof(yytext); }
[0-9]+             { num2 = atof(yytext); }
[+\-*/]            { op = yytext[0]; num1 = num2; }
\n                 {
                     switch(op) {
```

```
            case '+': result = num1 + num2; break;
            case '-': result = num1 - num2; break;
            case '*': result = num1 * num2; break;
            case '/': result = num1 / num2; break;
        }
        printf("The Answer = %f\n", result);
    }
%%
int main() {
    yylex();
    return 0;
}
int yywrap() {
    return 1;
}
```

Output:

```
deepayan@maverick-f15:/mnt/c/Users/Deepayan/Documents/MAVERICK/labs/5th-sem/cd-lab/lex-ep1$ flex p1.l
deepayan@maverick-f15:/mnt/c/Users/Deepayan/Documents/MAVERICK/labs/5th-sem/cd-lab/lex-ep1$ gcc lex.yy.c -o p1 -lfl
deepayan@maverick-f15:/mnt/c/Users/Deepayan/Documents/MAVERICK/labs/5th-sem/cd-lab/lex-ep1$ ./p1
15+14
The Answer = 29.000000
56*89
The Answer = 4984.000000
23-29
The Answer = -6.000000
9/4
The Answer = 2.250000
^C
deepayan@maverick-f15:/mnt/c/Users/Deepayan/Documents/MAVERICK/labs/5th-sem/cd-lab/lex-ep1$
```

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2. Write a program to match a string in LEX.

Code:

```
%{
#include <stdio.h>
int flag = 0;
%}
%%
```

```
"hello"    { flag = 1; }
.          { }
\n         {
              if (flag)
                 printf("String matched\n");
              else
                 printf("String not matched\n");
              flag = 0;
           }
%%

int main() {
   yylex();
   return 0;
}

int yywrap() {
   return 1;
}
```

Output:

```
deepayan@maverick-f15:/mnt/c/Users/Deepayan/Documents/MAVERICK/labs/5th-sem/cd-lab/lex-ep1$ nano p2.l
deepayan@maverick-f15:/mnt/c/Users/Deepayan/Documents/MAVERICK/labs/5th-sem/cd-lab/lex-ep1$ flex p2.l
deepayan@maverick-f15:/mnt/c/Users/Deepayan/Documents/MAVERICK/labs/5th-sem/cd-lab/lex-ep1$ gcc lex.yy.c -o p2 -lfl
deepayan@maverick-f15:/mnt/c/Users/Deepayan/Documents/MAVERICK/labs/5th-sem/cd-lab/lex-ep1$ ./p2
hello
String matched
^C
deepayan@maverick-f15:/mnt/c/Users/Deepayan/Documents/MAVERICK/labs/5th-sem/cd-lab/lex-ep1$
```

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3. Write a LEX program to count the number of vowels and consonants in a given string.

Code:

```
%{
#include <stdio.h>
```

```
int vowels = 0, consonants = 0;
%}

%%
[aeiouAEIOU]   { vowels++; }
[a-zA-Z]       { consonants++; }
.|\n           { }
%%

int main() {
    printf("Enter a string: ");
    yylex();
    printf("Number of vowels = %d\n", vowels);
    printf("Number of consonants = %d\n", consonants);
    return 0;
}

int yywrap() {
    return 1;
}
```

Output:

```
deepayan@maverick-f15:/mnt/c/Users/Deepayan/Documents/MAVERICK/labs/5th-sem/cd-lab/lex-ep1$ ./alpha
Enter a string: mynameisdeepayan
Number of vowels = 7
Number of consonants = 9
deepayan@maverick-f15:/mnt/c/Users/Deepayan/Documents/MAVERICK/labs/5th-sem/cd-lab/lex-ep1$
```